BFS Shell Stratification and the Emergence of Four-Dimensional Lorentzian Geometry

Q5b of the Cosmochrony Spectral Geometry Programme

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Abstract

The companion paper Q5a establishes, conditionally, that the admissibility forms $\mathcal{E}_q$ on the Weil representation of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ converge in the Mosco sense to a second-order operator $L_\Pi = -A\partial_x^2$ on $L^2(\mathbb{R})$, which is described as the "one-dimensional operator-theoretic shadow" of the homogeneous four-dimensional structure of $\mathrm{Heis}_3(\mathbb{R})$. The present paper completes the geometric reading deferred by Q5a. We establish three results. First, we show that the BFS shell stratification of $G_q = \mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ converges, in the pre-saturation regime, to the Carnot–Carathéodory sphere foliation of $\mathrm{Heis}_3(\mathbb{R})$; the homogeneous dimension $D_{\mathrm{hom}} = 4$ (Bass–Guivarc'h) endows the limiting geometry with the spectral and volume-growth properties of a four-dimensional space (Theorem 3.2). Second, under an explicit lifting hypothesis [H-lift], we identify L_Π as the image, under the Schrödinger representation, of the kinetic sector of the sub-Riemannian Laplacian Δ_H on $\mathrm{Heis}_3(\mathbb{R})$, and extract an effective contravariant co-metric tensor from the principal symbol $g^{\mu\nu} \propto A^{\mu\nu}$ from the principal symbol of the full effective operator (Theorem 5.2). Third, the Born–Infeld admissibility constraint forces this metric to carry Lorentzian signature $(-,+,+,+)$ (Theorem 6.1, following [3]). Each result is given with an explicit status: structural, conditional on [H-lift], or imported from [3].

Contents

1 Introduction

1.1 Context and what Q5a establishes

The Foundation paper [2] derives, from four axioms governing admissible non-injective transitions, that the symmetry group of the admissible fibre F_n is isomorphic to $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ for some prime q, and that F_n carries the Weil representation V_ρ on $L^2(\mathbb{Z}/q\mathbb{Z})$. The Cosmochrony white paper [4] identifies three qualitative convergences expected to hold in the large-q limit:

(i) The group $G_q = \mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ and its representation space $\mathcal{C}_q \simeq L^2(\mathbb{Z}/q\mathbb{Z})$ approach $\mathrm{Heis}_3(\mathbb{R})$ and $L^2(\mathbb{R})$.

(ii) The BFS shell structure of G_q approaches a continuous foliation, with polynomial ball growth $|B_n| \sim c\, n^4$ inducing a four-dimensional effective geometry.

(iii) The discrete admissibility filter Π_q approaches a second-order differential constraint, recovering the Born–Infeld Lagrangian in the continuum.

The companion paper Q5a [5] addresses convergences (i) and (iii). It establishes, conditionally on hypotheses [H1], [H2], [H-w], [H-E1] and Conjecture C, three results: a Hilbert inductive limit $\varinjlim_q \mathcal{C}_q \simeq L^2(\mathbb{R})$ (T1); convergence of rescaled Weil generators to metaplectic generators (T2); and Mosco convergence of the admissibility forms $\mathcal{E}_q$ to a second-order operator $L_\Pi = -A\partial_x^2$ on $L^2(\mathbb{R})$ (T3). Q5a states explicitly:

> "The operator L_Π is the one-dimensional operator-theoretic shadow of the homogeneous four-dimensional structure of $\mathrm{Heis}_3(\mathbb{R})$ (Bass–Guivarc'h, $D = 4$). The identification of L_Π with a four-dimensional effective geometry, the extraction of a metric tensor from its symbol, and the selection of the Lorentzian signature by the Born–Infeld admissibility condition are the subject of Q5b."

The present paper delivers this geometric reading. Convergence (ii) and the associated structural results are the primary content. The results combine classical geometric convergence theorems, structural consequences of the Cosmochrony framework, and conditional identifications stated explicitly as hypotheses; each result carries an unambiguous status label throughout.

1.2 Scope and main results

The paper is organised around three contributions:

Carnot convergence (Theorem 3.2; structural). We show that the BFS stratification $\{S_n\}$ of G_q converges, in the pre-saturation regime $n \ll q$, to the Carnot–Carathéodory sphere foliation of $\mathrm{Heis}_3(\mathbb{R})$. The homogeneous dimension $D_{\mathrm{hom}} = 4$ (Bass–Guivarc'h [1, 9]) gives the limiting geometry the spectral dimension of a four-dimensional space. This result is structural: it follows from the theory of asymptotic cones of nilpotent groups (Pansu [10]) and does not require any new hypothesis.

Metric extraction (Theorem 5.2; conditional on [H-lift]). Under Hypothesis [H-lift] (Definition 4.1), $L_\Pi = -A\partial_x^2$ is the image, under the Schrödinger representation, of the kinetic sector of the sub-Riemannian Laplacian $\Delta_H = \tilde{X}^2 + \tilde{Y}^2$ on $\mathrm{Heis}_3(\mathbb{R})$, restricted to the admissible sector. The full effective operator on the four-dimensional space $\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})$ has a principal symbol $A^{\mu\nu}k_\mu k_\nu$ from which the effective metric tensor $g^{\mu\nu} \propto A^{\mu\nu}$ is read off.

Lorentzian signature (Theorem 6.1; conditional on T5 and citing [3]). The Born–Infeld admissibility constraint forces $g^{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1)$ in homogeneous isotropic admissible regimes.

1.3 Notation and conventions

Throughout, q denotes a prime and $G_q = \mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ the discrete Heisenberg group. The BFS ball of radius n is $B_n = \{g \in G_q : d(e, g) \leq n\}$ and the BFS shell is $S_n = B_n \setminus B_{n-1}$. The sub-Riemannian Laplacian on $\mathrm{Heis}_3(\mathbb{R})$ is denoted Δ_H. The limit operator from Q5a is $L_\Pi = -A\partial_x^2$ acting on $L^2(\mathbb{R})$, where $A > 0$ is the admissibility weight from Q5a Hypothesis [H-w]. The principal symbol of a differential operator P of order m is $\sigma_m(P)(x, \xi)$. Status labels follow the convention of [5]: "proved" denotes a complete proof in the cited companion; "structural" denotes a consequence of classical results cited explicitly; "conditional" denotes a result following from a named hypothesis.

2 Sub-Riemannian Structure of $\mathrm{Heis}_3(\mathbb{R})$

2.1 The Heisenberg group as a Carnot group

The real Heisenberg group $\mathrm{Heis}_3(\mathbb{R})$ is the Lie group $\mathbb{R}^3$ equipped with the group law

$$(x, y, z) \cdot (x', y', z') = \big(x + x',\ y + y',\ z + z' + \tfrac{1}{2}(xy' - x'y)\big). \tag{1}$$

Its Lie algebra $\mathfrak{h}$ is spanned by left-invariant vector fields

$$\tilde{X} = \partial_x + \tfrac{y}{2}\,\partial_z, \qquad \tilde{Y} = \partial_y - \tfrac{x}{2}\,\partial_z, \qquad \tilde{Z} = \partial_z, \tag{2}$$

satisfying $[\tilde{X}, \tilde{Y}] = -\tilde{Z}$ and $[\tilde{X}, \tilde{Z}] = [\tilde{Y}, \tilde{Z}] = 0$. The Lie algebra admits a two-step stratification:

$$\mathfrak{h} = \mathfrak{g}_1 \oplus \mathfrak{g}_2, \qquad \mathfrak{g}_1 = \mathrm{span}\{\tilde{X}, \tilde{Y}\}, \qquad \mathfrak{g}_2 = \mathrm{span}\{\tilde{Z}\}. \tag{3}$$

The horizontal distribution $\mathcal{H} \subset T\mathrm{Heis}_3(\mathbb{R})$ is the left-invariant rank-2 distribution generated by $\mathfrak{g}_1$; it satisfies the Hörmander bracket condition $\mathfrak{g}_1 + [\mathfrak{g}_1, \mathfrak{g}_1] = \mathfrak{h}$, ensuring that any two points are connected by a horizontal curve (Chow–Rashevskii theorem). The central direction $\tilde{Z} = \partial_z$, generated by the commutator $[\tilde{X}, \tilde{Y}]$, is not horizontal; it spans the second layer $\mathfrak{g}_2$ and carries weight 2 under the anisotropic dilations (4). This weight-2 status is what makes the central direction sub-dominant in the horizontal geometry while contributing a full unit to the homogeneous dimension.

2.2 Anisotropic dilations and homogeneous dimension

The group $\mathrm{Heis}_3(\mathbb{R})$ admits a one-parameter family of anisotropic dilations

$$\delta_r(x, y, z) = (rx,\ ry,\ r^2 z), \qquad r > 0, \tag{4}$$

which are group homomorphisms and assign weights 1 to $\tilde{X}$ and $\tilde{Y}$, and weight 2 to $\tilde{Z}$. The *homogeneous dimension* is defined as

$$D_{\mathrm{hom}} = \sum_i \mathrm{weight}_i \cdot \dim(\mathfrak{g}_i) = 1 \cdot 2 + 2 \cdot 1 = 4. \tag{5}$$

Theorem 2.1 (Bass [1], Guivarc'h [9])**.** *For the discrete Heisenberg group* $\mathrm{Heis}_3(\mathbb{Z})$ *with the standard symmetric generating set* $\{X^{\pm 1}, Y^{\pm 1}\}$*, the BFS ball satisfies*

$$|B_n(\mathrm{Heis}_3(\mathbb{Z}))| \sim C\, n^4 \quad \text{as } n \to \infty, \tag{6}$$

for a constant $C > 0$*. The same growth law* $|B_n(G_q)| \sim C\, n^4$ *holds for* $G_q = \mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ *in the pre-saturation regime* $n \ll q$*.*

The exponent $4 = D_{\mathrm{hom}}$ is the Bass–Guivarc'h homogeneous dimension of $\mathrm{Heis}_3(\mathbb{R})$. A space with ball growth $|B_r| \sim r^D$ has Hausdorff dimension D with respect to the associated metric. Consequently, $(\mathrm{Heis}_3(\mathbb{R}), d_{\mathrm{CC}})$ has Hausdorff dimension 4, matching the spectral properties of a four-dimensional physical space (see Section 3).

2.3 Carnot–Carátheódory metric

Fix a left-invariant sub-Riemannian metric $g_{\mathcal{H}}$ on $\mathcal{H}$ making $\{\tilde{X}, \tilde{Y}\}$ orthonormal. The Carnot–Carathéodory distance between two points $p, q \in \mathrm{Heis}_3(\mathbb{R})$ is

$$d_{\mathrm{CC}}(p, q) = \inf_{\gamma} \int_0^1 \|\dot{\gamma}(t)\|_{\mathcal{H}}\, dt, \tag{7}$$

where the infimum is over all horizontal curves γ from p to q (curves whose velocity lies in $\mathcal{H}$ at every point). This metric is finite by Chow–Rashevskii and is left-invariant under the group action. The balls $B_r^{d_{\mathrm{CC}}}(e) = \{g \in \mathrm{Heis}_3(\mathbb{R}) : d_{\mathrm{CC}}(e, g) \le r\}$ satisfy $\mathrm{Vol}(B_r^{d_{\mathrm{CC}}}) \sim C\, r^4$ as $r \to \infty$, consistently with (5).

2.4 The sub-Riemannian Laplacian

The *Kohn–Laplacian* (sub-Laplacian) is the second-order differential operator

$$\Delta_H = \tilde{X}^2 + \tilde{Y}^2 = \left(\partial_x + \tfrac{y}{2}\partial_z\right)^2 + \left(\partial_y - \tfrac{x}{2}\partial_z\right)^2. \tag{8}$$

It is left-invariant, homogeneous of degree -2 under δ_r, and hypoelliptic (Hörmander's theorem). Its principal symbol at $p = (x, y, z)$ and covector $\xi = (\xi_x, \xi_y, \xi_z)$ is

$$\sigma_2(\Delta_H)(p, \xi) = \left(\xi_x + \tfrac{y}{2}\xi_z\right)^2 + \left(\xi_y - \tfrac{x}{2}\xi_z\right)^2. \tag{9}$$

This symbol is degenerate in the ξ_z direction (reflecting the sub-Riemannian structure) but is positive on horizontal covectors. The operator $-\Delta_H$ is essentially self-adjoint on $L^2(\mathrm{Heis}_3(\mathbb{R}))$ and its heat kernel satisfies

$$p_t(e, e) \sim C\, t^{-D_{\mathrm{hom}}/2} = C\, t^{-2} \quad \text{as } t \to 0^+, \tag{10}$$

which is the same decay rate as the heat kernel on $\mathbb{R}^4$. This is the precise sense in which $(\mathrm{Heis}_3(\mathbb{R}), \Delta_H)$ has *spectral dimension 4*.

3 BFS Stratification and Carnot Convergence

3.1 Asymptotic cone of the Heisenberg group

The following theorem is the cornerstone of the BFS-to-Carnot identification.

Theorem 3.1 (Pansu [10]). *Let $\Gamma = \mathrm{Heis}_3(\mathbb{Z})$ with word metric d_w associated to the generating set $S = \{X^{\pm 1}, Y^{\pm 1}\}$. For every $n \geq 1$, define the rescaled metric space $(\Gamma, n^{-1}d_w)$. As $n \to \infty$, this sequence of metric spaces converges in the Gromov–Hausdorff sense to $(\mathrm{Heis}_3(\mathbb{R}), d_{\mathrm{CC}})$. More precisely, for any $R > 0$ and $\varepsilon > 0$, there exists N_0 such that for all $n \geq N_0$, the ball $\{g \in \Gamma : d_w(e, g) \leq nR\}$ equipped with the metric $n^{-1}d_w$ is ε-close to $B_R^{d_{\mathrm{CC}}}(e)$ in the Gromov–Hausdorff distance.*

3.2 Application to the finite Heisenberg group

The finite group $G_q = \mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ differs from $\mathrm{Heis}_3(\mathbb{Z})$ only through the identification $k \equiv k+q$ in each coordinate direction. In the pre-saturation regime $n \ll q$, no such identification is active: the BFS ball $B_n(G_q)$ is isomorphic, as a metric ball, to $B_n(\mathrm{Heis}_3(\mathbb{Z}))$.

Theorem 3.2 (Carnot convergence; structural). *Fix $R > 0$. In the double limit $q \to \infty$, $n = \lfloor Rq^\alpha \rfloor$ with $\alpha < 1$, the rescaled BFS metric space $(G_q, n^{-1}d_w)$ restricted to balls of radius R converges in the Gromov–Hausdorff sense to the Carnot–Carathéodory ball $(B_R^{d_{\mathrm{CC}}}(e) \subset \mathrm{Heis}_3(\mathbb{R}), d_{\mathrm{CC}})$. The homogeneous dimension $D_{\mathrm{hom}} = 4$ gives the limiting space the volume growth and spectral dimension of a four-dimensional metric space.*

Proof. For $n \ll q$, the BFS ball $B_n(G_q) \cong B_n(\mathrm{Heis}_3(\mathbb{Z}))$ (no wraparound in any coordinate for $n < q/2$). This identification relies on the asymptotic cone description of nilpotent groups in the sense of Pansu [10] and on the Bass–Guivarc'h polynomial growth formula [1, 9]. Theorem 3.1 applied to $\Gamma = \mathrm{Heis}_3(\mathbb{Z})$ then gives the Gromov–Hausdorff convergence at scale n. The Hausdorff dimension and spectral dimension of $(\mathrm{Heis}_3(\mathbb{R}), d_{\mathrm{CC}})$ equal $D_{\mathrm{hom}} = 4$ by the Bass–Guivarc'h formula (5) and the heat-kernel asymptotics of Section 2. $\square$

3.3 Effective four-dimensional foliation

The BFS stratification $\{S_n\}_{n \geq 0}$ of G_q is a partition of G_q by "shells" at fixed word-metric distance from the identity. In the large-q limit, Theorem 3.2 identifies this stratification with the foliation of $(\mathrm{Heis}_3(\mathbb{R}), d_{\mathrm{CC}})$ by CC spheres $\partial B_r^{d_{\mathrm{CC}}}(e)$ at increasing radii $r = n/n_{\max}$.

Remark 3.3 (The four-dimensional reading). In Cosmochrony, the observable spacetime is the product $\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})$, where τ denotes the projective ordering index (BFS depth) from [7, 4]. As a smooth manifold, $\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R}) \cong \mathbb{R}^4$. The product structure is not postulated: it reflects the separation between the projective ordering direction τ (structural output of the admissibility cascade [7, 4]) and the relational spatial structure encoded by $\mathrm{Heis}_3(\mathbb{R})$ (structural output of the Heisenberg symmetry derivation of [2, 6]). The BFS shells $\{S_n\}$ foliate this product: for fixed $\tau = n$, each shell is a copy of the CC sphere $\partial B_n^{d_{\mathrm{CC}}}$ in the Heis_3 factor. The homogeneous dimension $D_{\mathrm{hom}} = 4$ of $\mathrm{Heis}_3(\mathbb{R})$ means that the spatial factor $\mathrm{Heis}_3(\mathbb{R})$ alone carries the volume growth and heat-kernel decay of a four-dimensional space. The full effective spacetime is therefore four-dimensional in the topological sense (as $\mathbb{R}^4$) and also carries a four-dimensional spectral structure in the spatial factor alone. This is the structural content of convergence (ii) from the Cosmochrony white paper [4].

Remark 3.4 (Relation to the O-series data). The O-series numerical campaign [5] confirms $D_{\mathrm{hom}} = 4$ empirically: the fitted ball-growth exponent $\hat{D}$ converges to 4 monotonically as q increases across primes $q \in \{11, \ldots, 401\}$ (O9–O25). The pre-saturation window of width $\Omega(q^{1/2})$ BFS steps (Q5a Window-Depth Theorem) ensures that the Pansu convergence of Theorem 3.2 is visible for all tested primes.

4 Lifting Hypothesis and Sub-Riemannian Operator

4.1 The Schrödinger representation and the 1D shadow

The sub-Laplacian Δ_H acts on functions $f(x, y, z)$ of three variables. Q5a, however, produces a *one-dimensional* operator $L_\Pi = -A\partial_x^2$ acting on $L^2(\mathbb{R})$. Bridging these two objects is the central open step of this paper: we identify the natural candidate for this bridge (the Schrödinger representation), formulate a precise hypothesis [H-lift] in Section 4.3, and give structural evidence; but we do not prove [H-lift] here.

The connection between Δ_H and one-dimensional operators runs through the Schrödinger representation.

For each $\xi \neq 0$, the Schrödinger representation $\pi_\xi : \mathrm{Heis}_3(\mathbb{R}) \to \mathcal{U}(L^2(\mathbb{R}))$ is defined by

$$(\pi_\xi(x, y, z)f)(k) = e^{2\pi i\xi(z+yk)} f(k + x), \qquad f \in L^2(\mathbb{R}). \tag{11}$$

The infinitesimal generators act as

$$d\pi_\xi(\tilde{X}) = \partial_k, \qquad d\pi_\xi(\tilde{Y}) = 2\pi i\xi k, \qquad d\pi_\xi(\tilde{Z}) = 2\pi i\xi \,\mathrm{Id}. \tag{12}$$

Under this representation, the sub-Laplacian becomes

$$d\pi_\xi(\Delta_H) = d\pi_\xi(\tilde{X})^2 + d\pi_\xi(\tilde{Y})^2 = \partial_k^2 - (2\pi\xi)^2 k^2. \tag{13}$$

This is the quantum harmonic oscillator (up to sign and scaling). The two terms correspond to:

- ∂_k^2: the *kinetic* sector, acting purely as $-(-\partial_k^2)$;
- $-(2\pi\xi)^2 k^2$: the *potential* sector, growing with k^2.

The operator $L_\Pi = -A\partial_x^2$ produced by Q5a corresponds to (a multiple of) the kinetic sector, in the variable $k \equiv x$.

4.2 Admissibility filter and sector dominance

The admissibility filter Π_q of Q5a selects a *low-energy* subspace of $\mathcal{C}_q = L^2(\mathbb{Z}/q\mathbb{Z})$. In the large-$q$ limit, this subspace consists of functions $f \in L^2(\mathbb{R})$ whose Fourier support is concentrated near $|\xi| \ll \xi_{\max} = q/2$. In this regime:

- The potential term $(2\pi\xi)^2 k^2$ contributes at energy scale $(2\pi\xi)^2\langle k^2\rangle$, which is suppressed relative to the kinetic term $\|\partial_k f\|^2$ for admissible functions whose k-spread is bounded by $A_n^{\max} = c_\chi/\sqrt{\lambda_n}$ (BI admissibility envelope).
- In the admissible sector, $\langle k^2\rangle \lesssim (c_\chi)^2/\lambda_{\min}$ is uniformly bounded, while the kinetic energy $\|\partial_k f\|^2$ grows with the number of independent Weil fingerprints.

The Mosco convergence result (T3 of Q5a) shows that the admissibility forms $\mathcal{E}_q(f, f) \to A\int |\partial_x f|^2\, dx$, capturing only the kinetic sector. The potential term is sub-leading in the admissible regime and does not contribute to the Mosco limit. This dominance is consistent with the coercivity of the Mosco limit form, which excludes oscillatory contributions from the potential sector: a coercive Dirichlet form must be positive-definite on its domain, whereas the potential term $-(2\pi\xi)^2 k^2$ is negative-definite and would destroy coercivity if retained. This behaviour is consistent with the Mosco convergence of the admissible forms (T3 of Q5a): Mosco convergence suppresses oscillatory contributions and selects the dominant coercive quadratic part, leaving precisely the kinetic sector as the unique coercive limit.

4.3 The lifting hypothesis

Hypothesis 4.1 ([H-lift)**.**] The operator $L_\Pi = -A\partial_x^2$ obtained in Q5a as the Mosco limit of the admissibility forms $\mathcal{E}_q$ is the image, under the Schrödinger representation π_1 at unit central character $\xi = 1$, of the kinetic sector of the sub-Laplacian Δ_H on $\mathrm{Heis}_3(\mathbb{R})$, restricted to the admissible (low-energy) subspace of $L^2(\mathbb{R})$. Equivalently, the full sub-Laplacian Δ_H on $\mathrm{Heis}_3(\mathbb{R})$ is the pre-image (under π_1) of the harmonic oscillator $\partial_k^2 - k^2$, and L_Π is its kinetic component in the admissible sector.

Remark 4.2 (Justification of [H-lift).] Hypothesis [H-lift] is supported by three structural observations:

1. Q5a (T2) establishes that the rescaled Weil generators $\hat{X}_q/\sqrt{q}$, $\hat{Y}_q/\sqrt{q}$ converge strongly to the metaplectic generators $d\pi_1(\tilde{X}) = \partial_k$ and $d\pi_1(\tilde{Y}) = ik$, which are precisely the generators entering Δ_H via (12).
2. The Mosco limit (T3 of Q5a) captures only the kinetic sector, consistently with the sector-dominance analysis of Section 4.2. This is a structural prediction: the admissibility filter must project out the potential term to produce a coercive (not oscillatory) Dirichlet form.
3. The operator identity $d\pi_\xi(\Delta_H) = \partial_k^2 - (2\pi\xi)^2 k^2$ shows that the kinetic sector of Δ_H and L_Π share the same functional form $-A\partial_k^2$ up to the overall constant A (which absorbs the central character ξ and the admissibility weight from [H-w] of Q5a).

The main open step is a direct proof that the Mosco limit of $\mathcal{E}_q$ coincides with the kinetic sector of $d\pi_1(\Delta_H)$ modulo the potential contribution, which is deferred to open problem Q5b-O1. This lifting should be understood as a representation-theoretic correspondence rather than a pointwise identification of operators: it asserts that L_Π and the kinetic sector of $d\pi_1(\Delta_H)$ define the same quadratic form on the admissible subspace of $L^2(\mathbb{R})$, not that they are equal as unbounded operators on all of $L^2(\mathbb{R})$. This correspondence is understood in a weak (spectral or representation-theoretic) sense rather than as a pointwise operator identity; in particular, the domains may differ outside the admissible subspace.

Remark 4.3 (Scope of the harmonic analysis). A full derivation of [H-lift] would require the Plancherel formula for $\mathrm{Heis}_3(\mathbb{R})$ (decomposing $L^2(\mathrm{Heis}_3(\mathbb{R}))$ as a direct integral of Schrödinger representations π_ξ, $\xi \neq 0$), together with a careful tracking of how the admissibility filter Π_q acts on each fiber ξ. This analysis is not developed here. The present paper formulates [H-lift] as a precise structural hypothesis and provides the evidence described in Remark 4.2; the complete harmonic analysis is deferred to future work.

5 Principal Symbol and Effective Metric Tensor

5.1 The full effective operator on $\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})$

Under [H-lift], the admissibility Dirichlet form in the continuum limit corresponds to the kinetic sector of Δ_H acting on the spatial factor $\mathrm{Heis}_3(\mathbb{R})$. The complete effective operator on the four-dimensional space $\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})$ incorporates both the projective ordering direction τ and the Heisenberg spatial directions:

$$L_{\mathrm{eff}} = -A_\tau \partial_\tau^2 + A_H \Delta_H, \tag{14}$$

where $A_\tau > 0$ is the effective temporal coefficient from the projective ordering structure [7] and $A_H > 0$ is the spatial coefficient from [H-lift] (related to the constant A in $L_\Pi = -A\partial_x^2$). The sign structure in (14) reflects the hyperbolic character forced by Born–Infeld admissibility (Section 6).

Remark 5.1 (Relation to the 1D shadow). On the spatial slice $\{\tau = \text{const}\}$, the operator L_{eff} reduces to $A_H\Delta_H$. In the Schrödinger representation at $\xi = 1$, the kinetic sector of $A_H\Delta_H$ maps to $A_H\partial_k^2$, which is $-L_\Pi = A\partial_x^2$ upon relabeling $k \to x$ and identifying $A_H = A$. This confirms the reading of L_Π as the "one-dimensional shadow" stated in Q5a.

5.2 Principal symbol and metric extraction

Let $p = (\tau, x, y, z) \in \mathbb{R}_\tau \times \text{Heis}_3(\mathbb{R})$ and $k = (k_\tau, k_x, k_y, k_z)$ be the dual (momentum) covector. The principal symbol of L_{eff} at p and k is

$$\sigma_2(L_{\text{eff}})(p,k) = -A_\tau k_\tau^2 + A_H\sigma_2(\Delta_H)(p, k_x, k_y, k_z), \tag{15}$$

where $\sigma_2(\Delta_H)$ is given by (9). In coordinates, this reads

$$\sigma_2(L_{\text{eff}})(p,k) = -A_\tau k_\tau^2 + A_H\Big[\Big(k_x + \tfrac{y}{2}k_z\Big)^2 + \Big(k_y - \tfrac{x}{2}k_z\Big)^2\Big]. \tag{16}$$

Following the operator–metric correspondence of [3] (see also [8]), the effective metric tensor $g^{\mu\nu}(p)$ is read off from the principal symbol by

$$\sigma_2(L_{\text{eff}})(p,k) = g^{\mu\nu}(p)\, k_\mu k_\nu. \tag{17}$$

This defines $g^{\mu\nu}$ as a symmetric tensor on $\mathbb{R}_\tau \times \text{Heis}_3(\mathbb{R})$. To spell out the identification: the sub-Laplacian $\Delta_H = \tilde{X}^2 + \tilde{Y}^2$ is a sum of squares of horizontal vector fields, hence its principal symbol

$$\sigma_2(\Delta_H)(p,\xi) = \sum_{i\in\{x,y\}} (\tilde{v}_i(p)\cdot\xi)^2, \qquad \tilde{v}_x = \tilde{X},\ \tilde{v}_y = \tilde{Y}, \tag{18}$$

defines a positive semi-definite quadratic form on $T^*\text{Heis}_3(\mathbb{R})$ whose matrix

$$A_H^{jk}(p) = \sum_{i\in\{x,y\}} [\tilde{v}_i(p)]^j [\tilde{v}_i(p)]^k$$

is the *horizontal co-metric tensor* of the sub-Riemannian structure. The full symbol of L_{eff} adds the temporal direction with a negative coefficient $-A_\tau$, consistently with the hyperbolic structure required in Section 6:

$$g^{\mu\nu}(p) = \text{diag}(-A_\tau) \oplus A_H \cdot A_H^{ij}(p). \tag{19}$$

This identification follows the standard correspondence between second-order differential operators and quadratic forms on the cotangent bundle [3, 8]. The metric $g^{\mu\nu}$ is not postulated: it is read off from the operator L_{eff} constructed from the admissibility data, under [H-lift]. The metric tensor obtained in this way is not a fundamental geometric object defined at the level of the substrate χ, but an effective structure arising from the admissible projection of spectral data onto the continuum limit; it carries no ontological status beyond this projection. This extraction is canonical once the effective operator L_{eff} is fixed: the principal symbol uniquely determines $g^{\mu\nu}$ up to the overall scale absorbed into the normalisation of c.

Theorem 5.2 (Metric extraction; conditional on [H-lift] and Q5a hypotheses)**.** *Under Hypothesis [H-lift] and the Q5a hypotheses* [H1], [H-w], [H-E1], [C], *the Mosco limit operator* $L_\Pi = -A\partial_x^2$ *lifts to the effective operator* L_{eff} *on* $\mathbb{R}_\tau \times \text{Heis}_3(\mathbb{R})$ *given by* (14)*. Its principal symbol defines, via* (17)*, a leading-order effective co-metric tensor*

$$g^{\mu\nu}(p) = \text{diag}(-A_\tau,\ A_H,\ A_H,\ 0_z\textit{-corrected}) \tag{20}$$

on $\mathbb{R}_\tau \times \text{Heis}_3(\mathbb{R})$*, where the z-component carries sub-Riemannian corrections from the Heisenberg structure. This tensor should be understood as the leading-order effective co-metric associated with* *the principal symbol; it becomes a non-degenerate metric only after incorporation of sub-principal corrections, as discussed in Remark 5.3 and open problem Q5b-O2. In the homogeneous isotropic regime* $(x, y) \approx 0$ *(near the identity of* $\mathrm{Heis}_3(\mathbb{R})$*), the off-diagonal terms in* (16) *vanish and the symbol simplifies to* $\sigma_2(L_{\text{eff}}) \approx -A_\tau k_\tau^2 + A_H(k_x^2 + k_y^2)$*, giving a three-dimensional pseudo-Riemannian symbol at leading order.*

Remark 5.3 (Status of the z-direction)*.* The sub-Riemannian symbol (9) is degenerate in k_z: the z-direction (central element $\tilde{Z}$) does not appear in the principal symbol of Δ_H at leading sub-Riemannian order. Its contribution enters via sub-principal terms and is responsible for the distinction between the topological dimension 3 of $\mathrm{Heis}_3(\mathbb{R})$ and its Hausdorff dimension $D_{\text{hom}} = 4$. A complete 4D Lorentzian metric on $\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})$ requires an extension of the symbol framework to incorporate the sub-Riemannian corrections from the z-direction; this is deferred to open problem Q5b-O2. The Lorentzian signature result (Section 6) applies to the non-degenerate part of the symbol and holds independently of this subtlety. Crucially, the effective four-dimensionality of the geometry arises from the homogeneous (Hausdorff and spectral) dimension $D_{\text{hom}} = 4$ of the Carnot structure, not from a non-degenerate Riemannian metric at the principal-symbol level; these two notions of dimension must be carefully distinguished. The central direction $\tilde{Z}$ does not carry an independent kinetic term at the sub-Riemannian level: its contribution to the principal symbol of Δ_H is sub-leading (see (9)). Its effective dynamical role emerges only after projection, where it couples to the ordering parameter τ and contributes to the Lorentzian structure through the sub-principal corrections addressed in open problem Q5b-O2.

Remark 5.4 (Nature of the effective four-dimensionality)*.* The effective four-dimensionality obtained in this paper is not that of a non-degenerate pseudo-Riemannian manifold at the principal-symbol level. It arises from the homogeneous dimension $D_{\text{hom}} = 4$ of the underlying Carnot structure, which governs both volume growth and spectral scaling, as established in Section 2. The Lorentzian metric extracted from the principal symbol is therefore an effective object, valid after projection of the discrete admissibility structure onto the admissible continuum configurations; it does not claim to be a smooth metric at every point of $\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})$ without further hypotheses. The transition from the degenerate sub-Riemannian symbol to a non-degenerate effective metric is expected to occur through sub-principal corrections to the operator L_{eff}, which lift the central direction $\tilde{Z}$ and generate a full-rank quadratic form at the effective level; this is the precise mathematical content of open problem Q5b-O2.

6 Lorentzian Signature from Born–Infeld Admissibility

6.1 The signature selection argument

The following theorem is a consequence of the Born–Infeld analysis of [3]. It is imported from that paper and adapted to the present geometric setting; no independent proof is given here. We state it for completeness and give the main lines of argument to make the paper self-contained.

Theorem 6.1 (Lorentzian signature; conditional on Theorem 5.2, citing [3])**.** *In homogeneous and quasi-isotropic admissible regimes, the Born–Infeld admissibility constraint forces the effective metric defined by* (17) *to have Lorentzian signature* $(-, +, +, +)$*.*

Proof sketch, following [3] Sections 5–6. The operator L_{eff} must define a well-posed initial value problem in the projective ordering variable τ; this is required by the structural stability of the admissibility cascade [7, 4]. The principal symbol $g^{\mu\nu}k_\mu k_\nu$ determines the propagation properties of effective modes in the continuum limit, so $g^{\mu\nu}$ must be compatible with hyperbolicity in the sense of Hörmander.

Four signature classes must be excluded:

1. **Riemannian** $(+,+,+,+)$: produces a positive-definite symbol, hence an elliptic operator. An elliptic operator in (τ, x, y, z) implies instantaneous propagation of perturbations across the relational network, contradicting the finite Born–Infeld capacity bound $A_n^{\max} = c_\chi/\sqrt{\lambda_n}$ [2, 3].

2. **Ultra-hyperbolic (two or more time-like directions)**: fails to admit a globally well-posed initial value problem and is excluded by spectral stability of the effective operator [3].

3. **Degenerate (rank < 4)**: excluded at the level of the effective metric after incorporation of sub-principal corrections. The leading sub-Riemannian principal symbol is degenerate in the central direction k_z (Remark 5.3), but the full effective operator, including sub-principal contributions from the Heisenberg commutator structure, is expected to yield a non-degenerate metric on $\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})$ (open problem Q5b-O2). The argument here applies to the non-degenerate block $(-A_\tau, A_H, A_H)$ from the temporal and horizontal directions; the z-direction contributes positive-definiteness once the sub-principal corrections are incorporated, and does not affect the signature count.

4. **Parabolic**: implies the existence of a characteristic direction in which the Cauchy problem is ill-posed, contradicting the monotonic projective ordering of [7].

The unique remaining signature compatible with (i) hyperbolicity of $\sigma_2(L_{\mathrm{eff}})$, (ii) a finite characteristic cone, (iii) monotonic projective ordering along τ, and (iv) the bounded Born–Infeld admissibility constraint, is the Lorentzian signature $(-,+,+,+)$. The detailed proof is established in [3] Sections 5–6 and depends only on the bounded-flux constraint and spectral stability, both of which are present in the Foundation axioms [2]. After normalising the maximal admissible propagation speed to $c = 1$, the effective metric is dynamically fixed to $g^{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1)$ in the homogeneous isotropic regime. This selection mechanism is intrinsic to the admissibility constraints and does not rely on any external causal or geometric assumption introduced independently of the framework. □

Remark 6.2 (Emergence, not postulation)*.* The Lorentzian metric is not introduced as a background structure. It is the unique pseudo-Riemannian structure compatible with: the Heisenberg group symmetry (giving $D_{\mathrm{hom}} = 4$), the projective ordering (giving the temporal direction τ), and the Born–Infeld admissibility bound (excluding all other signatures). Minkowski geometry emerges as the unique stable effective description of a maximally symmetric admissible regime. The emergence of the Lorentzian structure is thus not postulated but results from the admissibility constraints governing the projection of unresolved configurations onto observable degrees of freedom — the same non-injective projection that generates the Heisenberg symmetry group [2, 6] and the BFS shell structure [4]. This is the structural content of the claim in [4] that spacetime is a "large-q limit of the discrete observable space $\mathcal{O}$, with the four-dimensional structure inherited from the homogeneous dimension of G_q."

7 Status Table and Open Problems

7.1 Status table

The following list summarises the logical status of every result in this paper:

- **BFS → Carnot convergence** (Theorems 2.1–3.2): classical, following Bass–Guivarc'h [1, 9] and Pansu [10].

- **Operator lifting** [H-lift] (Hypothesis 4.1): explicit hypothesis, supported by structural evidence from Q5a T2 but not proved here.

- **Metric extraction** (Theorem 5.2): structural consequence of [H-lift] together with the Q5a hypotheses [H1], [H-w], [H-E1], [C].
- **Lorentzian signature** (Theorem 6.1): derived from the Born–Infeld admissibility analysis of [3], conditional on Theorem 5.2.

The detailed status of each item is recorded in the table below.

Label	Content	Status
Thm. 2.1	Ball growth $\lvert B_n\rvert \sim Cn^4$, $D_{\mathrm{hom}} = 4$	Proved (Bass, Guivarc'h)
$D_{\mathrm{hom}} = 4$	Effective 4D spectral geometry (Carnot)	Structural (Carnot geometry)
Thm. 3.1	GH convergence to Carnot CC metric	Proved (Pansu)
Thm. 3.2	BFS shells converge to CC spheres, $D_{\mathrm{hom}} = 4$	Structural (from above)
[H-lift]	L_Π is kinetic sector of $d\pi_1(\Delta_H)$	Hypothesis (Open: Q5b-O1)
Thm. 5.2	Metric tensor $g^{\mu\nu} \propto A^{\mu\nu}$ from symbol	Conditional on [H-lift] + Q5a
Thm. 6.1	Lorentzian signature $(-,+,+,+)$	Conditional on Thm. 5.2, citing [3]
Q5b-O1	Prove [H-lift] via Plancherel decomposition	Open
Q5b-O2	Extend metric to full 4×4 with z-direction	Open
Q5b-O3	Identify A_τ, A_H from admissibility data	Open

7.2 Open problems

Q5b-O1 (Proof of [H-lift] via Plancherel decomposition). Use the Plancherel formula for $\mathrm{Heis}_3(\mathbb{R})$, which decomposes $L^2(\mathrm{Heis}_3(\mathbb{R}))$ as a direct integral $\int_{\xi\neq 0}^{\oplus} \mathcal{H}_\xi \, |\xi| \, d\xi$ of Schrödinger representation spaces $\mathcal{H}_\xi \cong L^2(\mathbb{R})$, to track how the admissibility filter Π_q acts on each fiber. The key step is showing that Π_q concentrates admissible modes on the kinetic sector ∂_k^2 of $d\pi_\xi(\Delta_H)$, i.e., that the potential energy $\|(2\pi\xi k)f\|^2$ of admissible functions $f = \Pi_q g$ is bounded by the BI capacity envelope, making the kinetic sector dominant in the Mosco limit. Expected difficulty: moderate, given T2 of Q5a.

Q5b-O2 (Extension to a full 4×4 Lorentzian metric). Theorem 5.2 and the signature argument of Theorem 6.1 concern only the non-degenerate part of the sub-Riemannian symbol. A complete treatment requires incorporating the z-direction, which enters the effective metric through sub-principal corrections. The natural framework is the theory of hypoelliptic operators and Beals–Greiner calculus, or alternatively the tangent groupoid construction for sub-Riemannian spaces. A full 4×4 Lorentzian metric on $\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})$ would require showing that the z-correction is positive-definite and contributes a third spatial direction to the effective geometry. Expected difficulty: hard; requires analysis of the sub-principal symbol of L_{eff}.

Q5b-O3 (Identification of A_τ and A_H from admissibility data). Theorem 5.2 introduces two positive constants A_τ and A_H characterising the temporal and spatial sectors of L_{eff}. Their identification from the admissibility data would require: expressing A_H in terms of the Born–Infeld saturation constant c_χ and the spectral weight A of Q5a (see Q5a-O5); and expressing A_τ in terms of the projective temporal ordering data from [7]. This identification is the first step toward an explicit metric from first principles and connects naturally to the Schwarzschild exterior solution derived in [3] Section 7.

The emergence of Lorentzian geometry established in this paper is thus traced back entirely to admissibility constraints on projected spectral data, rather than to any fundamental geometric postulate; spacetime structure is a derived consequence of the non-injective projection, not an input.

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